

Supporting Information for

Discovery of Wall-Selective Carbon Nanotube Growth Conditions *via*
Automated Experimentation

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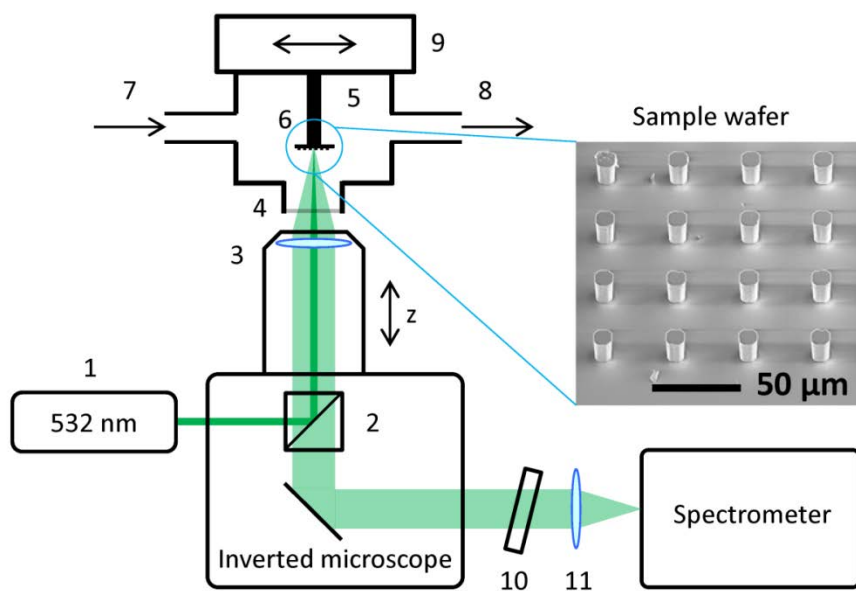


Figure S1. ARES instrument diagram. 1. Laser. 2. Beam splitter. 3. Objective lens. 4. Quartz window. 5. Growth chamber. 6. Sample wafer holder. 7. Gas in from mass flow controllers and water source (not shown). 8. Gas out to vacuum pump (not shown). 9. x-y-z motion stage. 10. Notch filter. 11. Lens. Inset shows SEM image of the pillars.

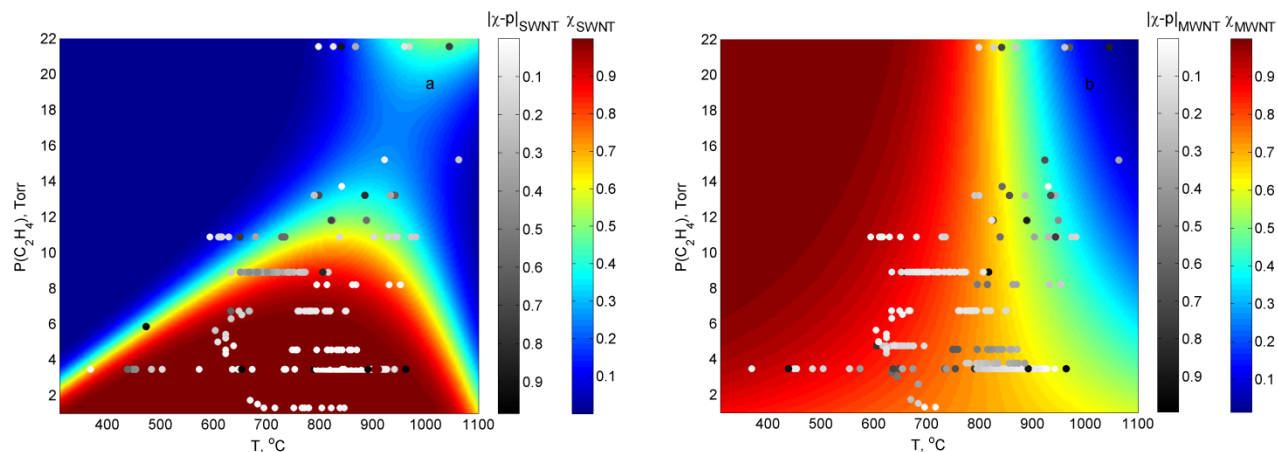


Figure S2. Comparison of experimental outcomes with predictions of logistic regression models.

Experiments are mapped as gray scale dots, and are superimposed for convenience onto 2-dimensional slices through 4-dimensional parameter space of the models predicting probabilities of nucleating SWNT (a) and MWNT (b) (“jet” color maps, similar to Figure 4), with other inputs fixed at typical average values: $P(H_2) = 12$ Torr and $H_2O = 50$ ppm.

We found that the best way to demonstrate direct connection between calculations and experimental data is to color dots with the difference between the model prediction χ and experimental outcome p , which is the figure of merit of the model. Therefore the dot color is proportional to $|\chi - p|_{SWNT}$ (a) and $|\chi - p|_{MWNT}$ (b), *i.e.* white dots represent experiments where experimental outcome agrees with the model prediction, while black dots represent experiments where model deviates from experiment.

(Experiments are mapped by projecting experimental parameters from 4-dimensional parameter space onto the 2-dimensional slice, *i.e.* each dot has two more experimental parameters ($P(H_2)$ and H_2O concentration) associated with it that do not necessarily belong on the slice. Because of that, coloring dots with experimental outcome (*e.g.* white for SWNT and black for MWNT) does not show correlation between experiment and models predictions.)

It can be seen that models are in a good overall agreement with outcomes of experiments (majority of light-colored dots). It can also be seen that sampling of the parameter space is far from even.

Many experiments fall onto horizontal lines on these maps. This happens when 25-experiment series are executed with the same C_2H_4 partial pressure while temperature is varied.

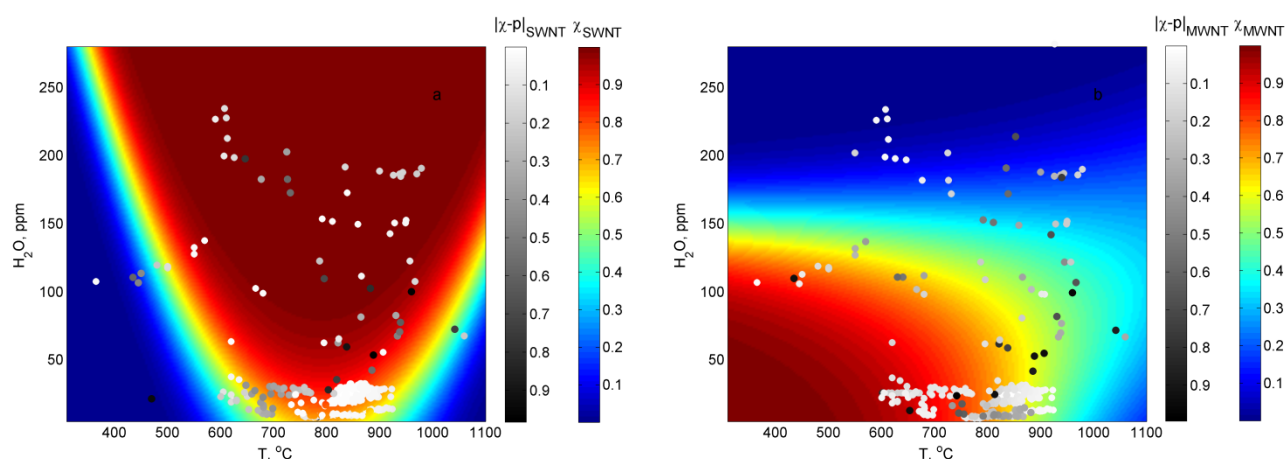


Figure S3. Comparison of experimental outcomes with predictions of logistic regression models. Experiments are mapped as gray scale dots, and are superimposed for convenience onto 2-dimensional slices through 4-dimensional parameter space of the models predicting probabilities of nucleating SWNT (a) and MWNT (b) (“jet” color maps, similar to Figure 7), with other inputs fixed at typical average values: $P(H_2) = 12$ Torr and $P(C_2H_4) = 8$ Torr.

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(a) and $|\chi-p|_{\text{MWNT}}$ (b), *i.e.* white dots represent experiments where experimental outcome agrees with the model prediction, while black dots represent experiments where model deviates from experiment.

(Experiments are mapped by projecting experimental parameters from 4-dimensional parameter space onto the 2-dimensional slice, *i.e.* each dot has two more experimental parameters ($P(\text{H}_2)$ and H_2O concentration) associated with it that do not necessarily belong on the slice. Because of that, coloring dots with experimental outcome (*e.g.* white for SWNT and black for MWNT) does not show correlation between experiment and models predictions.)

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